Module1:

1. Explain in your own words what a program is and how it functions.
   A program is a set of functions and rules that helps user understand the binary language and tells computer how to perform specific tasks.

2. What are the key steps involved in the programming process?
   Key steps in programming process are analysis, coding, design, debugging, testing.

3. What are the main differences between high-level and low-level programming languages?
   Difference between high-level and low-level is that low-level programming helps with performance, precision where as high-level programming helps with user readability.

4. Describe the roles of the client and server in web communication.
   The role of the client is to provide instruction where as a server's role is to follow the user instruction and provide with satisfactory result.

5. Explain the function of the TCP/IP model and its layers.
   TCP/IP manages network communication by transmitting data from software to physical device. It handles application-level data and ensures end to end transport of data via IP addresses.
   Its layers are Process, Host to Host, Network and Link.

6. Explain Client Server Communication
   Client server communication is type of process where user requests for data from browsers or applications to the servers.

7. How does broadband differ from fiber-optic internet?
   Broadband is a general term for high-speed internet (cable, satellite), while fiber-optic is a specific, superior broadband technology using light through glass strands.

8. What are the differences between HTTP and HTTPS protocols?

. HTTPS uses TLS (SSL) to encrypt communications, protecting sensitive user data (like passwords and credit cards) from eavesdropping and tampering, whereas HTTP transmits data in plain text, leaving it vulnerable.

9. What is the role of encryption in securing applications?
   Encryption acts as a primary defense in application security by converting sensitive data into unreadable cipher text, ensuring only authorized parties with the correct key can access it. Its basically a security officer of your data.

10. What is the difference between system software and application software?
    System software works in the background, while application software operates in the foreground upon user request

11. What is the significance of modularity in software architecture?
    Modularity in software architecture is the practice of improving code maintainability, reusability, and scalability, allowing developers to test, upgrade, or fix individual components without impacting the entire system's functionality.

12. Why are layers important in software architecture?
    Layers are important in software architecture because they promote modularity, separation of concerns, and maintainability by dividing an application into distinct parts with specific responsibilities.

13. Explain the importance of a development environment in software production.
    It provides safe, standardised, and efficient workspace to programmers for better results.

14. What is the difference between source code and machine code?
    The difference is that source code is human readable where as Machine code is only system readable because a machine can only understand binary language.

15. Why is version control important in software development?
    Version control systems (VCS) are important in software development for several key reasons: -
    Collaboration and Teamwork,
    -Tracking Changes and History,
    -Rollbacks and Error Recovery,

-Branching and Experimentation,
-Code Quality and Review,
-Backup and Disaster Recovery,
-Project Management and Organization.

16. What are the benefits of using Github for students?
    The benefits of using Github is that its like a whole market but for programmers and and students wanting to be programmers. It is a collection of all the study, practice materials and not just for students it also a gold mine for professional developers too if they get stuck or need help with specific topics.

17. What are the differences between open-source and proprietary software?
    The difference between open-source and proprietary software is that, Open Source is publicly available, anyone can use it to learn or re-write and modify and re-distribute it, where-as proprietary software is only owned by company and can only be used by authorised users restricted by licence.

18. How does GIT improve collaboration in a software development team?
    Git significantly improves collaboration in a software development team through several key features and a distributed workflow:
    -Parallel Development with Branching
    -Efficient Merging and Conflict Resolution
    -Code Review and Feedback with Pull Requests
    -Clear Version History and Accountability
    -Distributed Nature.

19. What is the role of application software in businesses?
    It helps business smooth their operations by providing better communication, all data stored at one place and collaboration.

20. What are the main stages of the software development process?
    Main stages of software development are:
    -Coding
    -Designing
    -Debugging
    -Execution

21. Why is the requirement analysis phase critical in software development?
    It ensures that the software is solving the problem which it was build to solve, so that it can provide better user with better final product without any bugs and errors.

22. What are the key elements of system design?
    The key elements of system design are architecture, components, modules and interfaces.

23. Why is software testing important?
    Software Testing is important because it finds all the flaws and can be fixed before making it online to the user to provide the best service without any operation failure or bug issues.

24. What types of software maintenance are there?
    There are 4 types of maintenance
    -Corrective (finding bugs)
    -Adaptive (Adapting the system)
    -Perfective (Ensuring smooth operation)
    -Preventive (Preventing major shutdown after its made public)

25. What are the advantages of using web applications over desktop applications?
    The advantages of using web applications over desktop applications are that it does not occupy any system space, every time user access the web app its provided with the latest version and its independent as it does not run on the hardware system rather it runs on the browser.

26. What are the key differences between web and desktop applications?
    The difference between web and desktop application is that web application does not need any installation as it runs on browser but it do require internet where as desktop application can be accessed offline.

27. What role does UI/UX design play in application development?
    UI/UX design plays a very important role, it makes sure that the app is visually pleasing and user friendly at the same time.

28. What are the differences between native and hybrid mobile apps?
    Difference between Native and Hybrid mobile apps is that Native is build to use on single device where as hybrid apps can be used on multiple platforms.

29. What is the significance of DFDs in system analysis?
    DFD's provide a better understanding of how the data moves through the system and makes complex processes easy to understand, It also improves communication between the system and the user.

30. What are the pros and cons of desktop applications compared to web applications?
    Desktop applications excel with better performance and offline access, but they require installation and lack cross-platform compatibility.
    -Web applications are accessible from anywhere with an internet connection and are easy to update, but they often have slower performance and require an internet connection to function.

31. How do flowcharts help in programming and system design?
    Flowcharts help in programming and system design by providing a visual blueprint to map out processes, algorithms, and system structures, which aids in planning, debugging, and communication.